BS EN 50600-1:2019

BSI Standards Publication

Information technology - Data centre facilities and infrastructures

Part 1: General concepts

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National foreword

This British Standard is the UK implementation of EN 50600-1:2019. It supersedes BS EN 50600-1:2012, which is withdrawn.

The UK participation in its preparation was entrusted to Technical Committee TCT/7, Telecommunications - Installation requirements.

A list of organizations represented on this committee can be obtained on request to its secretary.

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European foreword

This document (EN 50600-1:2019) has been prepared by CLC/TC 215 “Electrotechnical aspects of telecommunication equipment”.

The following dates are fixed:

- latest date by which this document has to be implemented at national level by publication of an identical national standard or by endorsement (dop) 2020–04–29
- latest date by which the national standards conflicting with this document have to be withdrawn (dow) 2022–04–29

This document supersedes EN 50600-1:2012.

The following major modifications have been made compared to EN 50600-1:2012:

a) reference to Key Performance Indicators of EN 50600-4-X included;

b) Clause 7 (Availability) has been revised;

c) the design processes (Clause 8) and design principles (Clause 9) have been moved from an annex to the main body of the document;

d) existing Annex A has been removed;

e) new Annexes A and B have been added.

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Introduction

The unrestricted access to internet-based information demanded by the information society has led to an exponential growth of both internet traffic and the volume of stored/retrieved data. Data centres are housing and supporting the information technology and network telecommunications equipment for data processing, data storage and data transport. They are required both by network operators (delivering those services to customer premises) and by enterprises within those customer premises.

Data centres usually need to provide modular, scalable and flexible facilities and infrastructures to easily accommodate the rapidly changing requirements of the market. In addition, energy consumption of data centres has become critical both from an environmental point of view (reduction of environmental footprint) and with respect to economical considerations (cost of energy) for the data centre operator.

The implementation of data centres varies in terms of:

a) purpose (enterprise, co-location, co-hosting or network operator facilities);

b) security level;

c) physical size;

d) accommodation (mobile, temporary and permanent constructions).

The needs of data centres also vary in terms of availability of service, the provision of security and the objectives for energy efficiency. These needs and objectives influence the design of data centres in terms of building construction, power distribution, environmental control, telecommunications cabling and physical security as well as the operation of the data centre. Effective management and operational information is required to monitor achievement of the defined needs and objectives.

Recognizing the substantial resource consumption, particularly of energy, of larger data centres, it is also important to provide tools for the assessment of that consumption both in terms of overall value and of source mix and to provide Key Performance Indicators (KPIs) to evaluate trends and drive performance improvements.

At the time of publication of this European Standard, EN 50600 series is designed as a framework of standards and technical reports covering the design, the operation and management as well as the key performance indicators for energy efficient operation of the data centre.

The EN 50600-2 series defines the requirements for the data centre design.

The EN 50600-3 series defines the requirements for the operation and the management of the data centre.

The EN 50600-4 series defines the key performance indicators for the data centre.

The CLC/TR 50600-99-X Technical Reports cover recommended practices and guidance for specific topics around data centre operation and design.

This series of European Standards specifies requirements and recommendations to support the various parties involved in the design, planning, procurement, integration, installation, operation and maintenance of facilities and infrastructures within data centres. These parties include:

1) owners, operators, facility managers, ICT managers, project managers, main contractors;

2) consulting engineers, architects, building designers and builders, system and installation designers, auditors, test and commissioning agents;

3) facility and infrastructure integrators, suppliers of equipment;

4) installers, maintainers.

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The inter-relationship of the standards and technical reports within the EN 50600 series is shown in Figure 1.

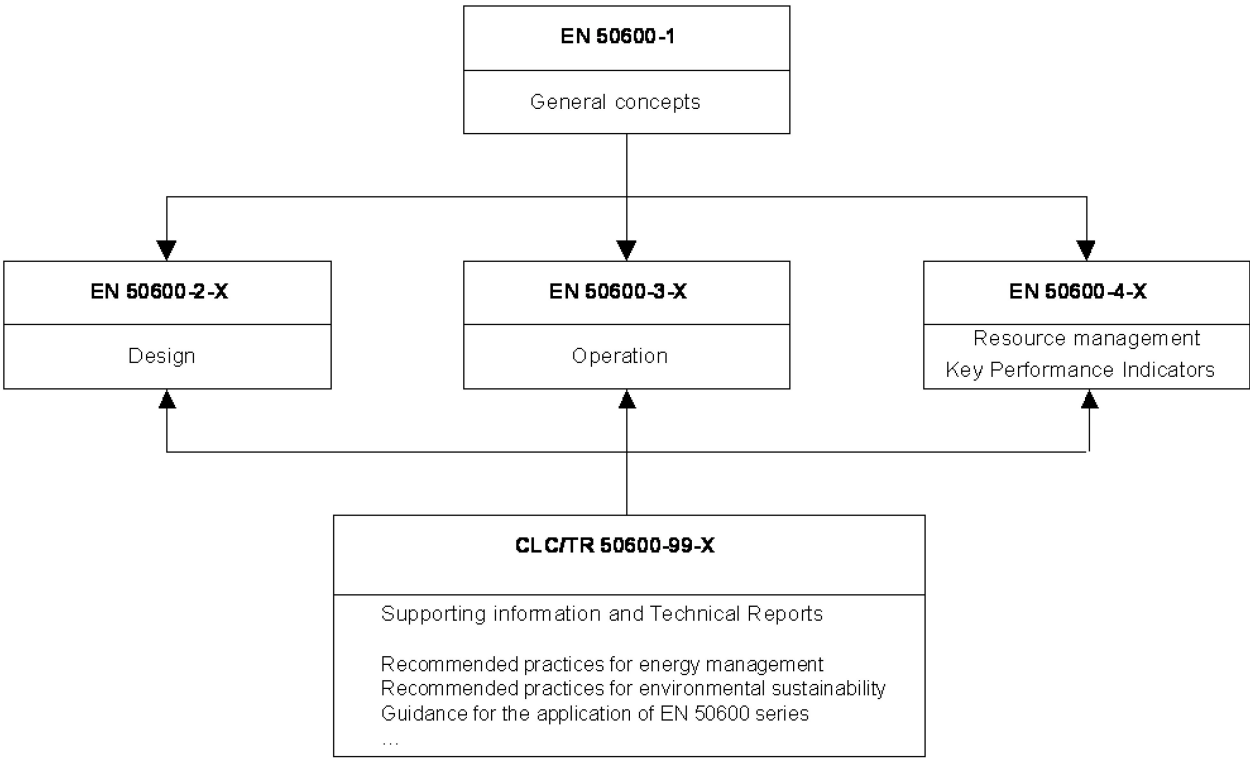

Figure 1 – Schematic relationship between EN 50600 series of standards

This European Standard specifies general requirements for data centres for all kinds of data centres irrespective of their size and physical construction. It introduces a classification system for availability, physical security and energy efficiency enablement.

EN 50600-2-X standards specify requirements and recommendations for particular facilities and infrastructures to support the relevant classification for “availability”, “physical security” and “energy efficiency enablement” selected from EN 50600-1.

EN 50600-3-X documents specify requirements and recommendations for data centre operations, processes and management.

EN 50600-4-X documents specify requirements and recommendations for key performance indicators (KPIs) used to assess and improve the resource usage efficiency and effectiveness, respectively, of a data centre.

This European Standard is intended for use by and collaboration between architects, building designers and builders, system and installation designers.

This series of European Standards does not address the selection of information technology and network telecommunications equipment, software and associated configuration issues.

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1 Scope

This document:

a) describes the general principles for data centres upon which the requirements of the EN 50600 series are based;

b) defines the common aspects of data centres including terminology, parameters and reference models (functional elements and their accommodation) addressing both the size and complexity of their intended purpose;

c) describes general aspects of the facilities and infrastructures required to support data centres;

d) specifies a classification system, based upon the key criteria of “availability”, “security” and “energy-efficiency” over the planned lifetime of the data centre, for the provision of effective facilities and infrastructure;

e) details the issues to be addressed in a business risk and operating cost analysis enabling application of the classification of the data centre;

f) provides reference to operation and management of data centres;

g) introduces the concepts of Key Performance Indicators (KPIs) for resource management of data centre facilities and infrastructures.

The following topics are outside of the scope of this series of European Standards:

1) the selection of information technology and network telecommunications equipment, software and associated configuration issues are outside the scope of this European Standard;

2) quantitative analysis of overall service availability resulting from multi-site data centres;

3) safety and electromagnetic compatibility (EMC) requirements (covered by other standards and regulations. However, information given in this European Standard can be of assistance in meeting these standards and regulations).

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

EN 50600-2-1, *Information technology - Data centre facilities and infrastructures - Part 2-1: Building construction*

EN 50600-2-2, *Information technology - Data centre facilities and infrastructures - Part 2-2: Power supply and distribution*

EN 50600-2-3, *Information technology - Data centre facilities and infrastructures - Part 2-3: Environmental control*

EN 50600-2-4, *Information technology - Data centre facilities and infrastructures - Part 2-4: Telecommunications cabling infrastructure*

EN 50600-2-5, *Information technology - Data centre facilities and infrastructures - Part 2-5: Security systems*

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EN 50600-3-1, *Information technology - Data centre facilities and infrastructures - Part 3-1: Management and operational information*

EN 50600-4-X (all parts), Information technology – Data centre facilities and infrastructures – Part 4-X

3 Terms, definitions and abbreviations

3.1 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

• IEC Electropedia: available at http://www.electropedia.org/

• ISO Online browsing platform: available at http://www.iso.org/obp

3.1.1
availability
ability of an item to be in a state to perform a required function under given conditions at a given instant of time or over a given time interval, assuming that the required external resources are provided

[SOURCE: IEC 60050-191:1990, 191-02-05]

3.1.2
building entrance facility
facility that provides all necessary mechanical and electrical services and which complies with all relevant regulations for the entry of specific services or infrastructures into a building

[SOURCE: EN 50173-1:2018, 3.1.18 – modified: replaced “telecommunication cables” with “specific infrastructures or services” and deleted „and which can enable transmission from outdoor to indoor cable”]

3.1.3
building security
facilities and systems necessary to provide the required levels of security at the entrance to and within the building containing the data centre

3.1.4
cabinet
enclosed construction for housing closures and other information technology equipment

[SOURCE: EN 50174-1:2018, 3.1.7]

3.1.5
co-hosting data centre
data centre in which multiple customers are provided with access to network(s), servers and storage equipment on which they operate their own services/applications

Note 1 to entry: Both the information technology equipment and the support infrastructure of the building are provided as a service by the data centre operator.

[SOURCE: EN 50174-2:2018, 3.1.2]

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3.1.6
co-location data centre
data centre in which multiple customers locate their own network(s), servers and storage equipment

Note 1 to entry: The support infrastructure of the building (such as power distribution and environmental control) is provided as a service by the data centre operator.

[SOURCE: EN 50174-2:2018, 3.1.3]

3.1.7
computer room space
area within the data centre that accommodates the data processing, data storage and telecommunication equipment that provides the primary function of the data centre

3.1.8
control room space
area within the data centre used to control the operation of the data centre and to act as a central point for all control and monitoring functions

3.1.9
data centre
structure, or group of structures, dedicated to the centralised accommodation, interconnection and operation of information technology and network telecommunications equipment providing data storage, processing and transport services together with all the facilities and infrastructures for power distribution and environmental control together with the necessary levels of resilience and security required to provide the desired service availability

Note 1 to entry: A structure can consist of multiple buildings and/or spaces with specific functions to support the primary function.

Note 2 to entry: The boundaries of the structure or space considered the data centre which includes the information and communication technology equipment and supporting environmental controls can be defined within a larger structure or building.

3.1.10
data centre security
facilities and systems that provide the required levels of security at the entrance to and within the data centre

3.1.11
demarcation point
point where the operational control or ownership changes

3.1.12
electrical distribution space
area used for housing facilities to distribute electrical power between the transformer space and electrical spaces within the data centre or elsewhere within the premises or individual buildings within the premises

3.1.13
electrical space
area within the data centre used for housing facilities to deliver and control electrical power to the data centre spaces (including switchboards, batteries, uninterruptible power systems (UPS) etc.)

3.1.14
enterprise data centre
data centre that is operated by an enterprise which has the sole purpose of the delivery and management of services to its employees and customers

[SOURCE: EN 50174-2:2018, 3.1.8]

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3.1.15
external premises security
facilities and systems that provide the required levels of security for the area between the building and the boundary of the premises

3.1.16
energy efficiency enablement
ability to measure the energy consumption and to allow calculation and reporting of energy efficiency of the various facilities and infrastructures

3.1.17
facility
spaces and pathways that accommodate a specific infrastructure

3.1.18
functional capability
ability of the data centre (or system or subsystem) to deliver its intended function

3.1.19
functional element
source of supply, device or path

3.1.20
generator space
area used for housing the installation of electrical power supply generation equipment together with control systems, storage of associated fuels or energy conversion equipment

3.1.21
holding space
area within the data centre used for the holding of equipment prior to being brought into service or having been taken out of service

3.1.22
infrastructure
technical systems providing functional capability of the data centre (e.g. power distribution, environmental control and physical security)

3.1.23
main distributor
distributor used to make connections between the main distribution cabling subsystem, network access cabling subsystem and active equipment

[SOURCE: EN 50173-5:2018, 3.1.10]

3.1.24
mechanical space
area that is used for housing mechanical equipment and infrastructure that provides environmental control for the data centre spaces (including chillers and water treatment, air handling and fire suppression systems)

3.1.25
network operator data centre
data centre that has the primary purpose of the delivery and management of broadband services to the operators' customers

[SOURCE: EN 50174-2:2018, 3.1.18]

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3.1.26
physical security
active and passive measures (combining physical and technological controls), procedures and responsibilities to maintain the desired level of availability for the facilities and infrastructures of the data centres in relation to access control and environmental events

3.1.27
planned downtime
period of time during which a system or subsystem does not provide functional capability whilst it undergoes maintenance or is switched off to test the response of a related system or subsystem

3.1.28
premises entrance facility
space that provides all necessary mechanical and electrical services for the entry of cables into the premises

3.1.29
storage space
area where general goods and/or data centre goods to be used in the premises and data centre are stored

3.1.30
system
set of interrelated functional elements considered in a defined context as a whole and separated from their environment

3.1.31
telecommunications
technology concerned with the transmission, emission, and reception of signs, signals, writings, images, and sounds, by cable, radio, optical, or other electromagnetic systems

Note 1 to entry: The term telecommunications has no legal meaning when used in this document

[SOURCE: EN 50173-1:2018, 3.1.49]

3.1.32
telecommunications cabling
telecommunications cabling infrastructure from the telecommunications space(s) to the premises entrance facility

3.1.33
telecommunication equipment
equipment within the data centre that provides telecommunication services within the data centre

3.1.34
telecommunications space
area which may house demarcation points and telecommunication equipment associated with the building entrance facility and which may allow service providers restricted access to the data centre

3.1.35
testing space
area within the data centre used for the testing and configuring of equipment prior to being brought into service

Note 1 to entry: Testing space is sometimes called staging area.

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3.1.36
transformer space
area used for housing equipment necessary to convert voltage levels and/or provide necessary isolation for the connection to the equipment within the premises or individual buildings within the premises

3.1.37
uninterruptible power system
combination of convertors, switches and energy storage devices (such as batteries), constituting a power system for maintaining continuity of load power in case of input power failure

Note 1 to entry: Continuity of load power occurs when voltage and frequency are within rated steady-state and transient tolerance bands and with distortion and interruptions within the limits specified for the output port. Input power failure occurs when voltage and frequency are outside rated steady-state and transient tolerance bands or with distortion or interruptions outside the limits specified for the UPS.

[SOURCE: EN 62040-1:2017, 3.101]

3.1.38
unplanned downtime
time taken, following a failure of functional capability, to repair the relevant infrastructure together with the "re-boot" time necessary to recover functional capability following that repair

3.2 Abbreviations

For the purposes of this document the following abbreviations apply:

CRAC	Computer Room Air Conditioning (Unit)
CRAH	Computer Room Air Handler units
EMI	electromagnetic interference
ffs	for further study
MTBF	Mean Time Between Failures
MTTR	Mean Time To Repair
NOC	Network Operating Centre
UPS	Uninterruptible Power System

4 Conformance

For a data centre design to conform to this European Standard:

a) a business risk analysis according to Clause 5 shall be completed;

b) an appropriate Availability Class in 7.2 shall be selected using a business risk analysis in Clause 5;

c) appropriate Protection Classes for the data centre pathways and spaces shall be in accordance with 7.3.1;

d) an appropriate energy efficiency enablement level in 7.4 shall be selected;

e) the design process of Clause 8 (or equivalent) shall be applied;

f) the design principles of Clause 9 shall be applied.

NOTE The application of the design process in Clause 8 is not mandatory for an assessment of existing data centres.

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5 Business risk analysis

5.1 General

The overall availability of a data centre is a measure of the continuity of its data processing, storage, and transport functions. The acceptable level of the overall availability of a data centre is determined by a number of factors including:

a) a business impact analysis (see 5.2) - the cost associated with a failure of service provision, which depends upon a number of factors including the function and importance of the data centre;

b) externally applied commercial pressures (e.g. insurance costs).

There is a link between the availability of the infrastructures specified in EN 50600-2-X standards and the overall availability but it should be recognized that the recovery of intended data processing, storage, and transport functionality following the repair of an infrastructure failure depends on many factors related to the configuration of the hardware and software providing that functionality.

As a result, the role of the infrastructure is to support overall availability objectives but is not the sole factor in their attainment.

The availability of each of the facilities and infrastructures of the data centre required to support the desired overall availability is described by an availability classification (see 7.2). The design of each of the data centre infrastructures shall take account of their impact on overall availability and the costs associated with the predicted downtime associated with failure or planned maintenance.

The design and physical security of the facilities and infrastructures of the data centre shall be subjected to a risk analysis (see 5.3) which maps identified risk events against the requirements of the availability classification (see 7.2). The availability classification for each infrastructure is described as providing low, medium, high and very high availability. Clause 7 further describes the situations (risk events) for which each infrastructure is protected against failure. Other approaches are to apply “% availability” to infrastructures but this is not supported by this standard series for reasons explained in Annex A.

A business risk analysis identifies the aspects of the facilities and infrastructures that require investment in terms of design improvements to reduce their impact and/or probability of those risk events.

5.2 Business impact analysis

This standard does not define methods of analysis for the cost of downtime. Standards such as EN 31010, ISO/TS 22317 or EN ISO 22301 provide useful guidance.

The parameters to be considered within such an analysis will depend upon the purpose of the data centre. Some organisations can assign a monetary value (or range) to loss of service which may include the following:

a) immediate financial penalties;

b) consequential losses;

c) an assessment of longer term damage to business reputation e.g. an Internet Service Provider or a financial institution.

Although cost is often considered when analysing downtime, other impacts should also be considered. Data centres containing life safety, legal, medical and criminal information may have individually recognized consequences from un-scheduled downtime.

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5.3 Risk analysis

This standard does not define methods of risk analysis. Standards such as EN 31010 provide useful guidance.

Risk analysis may be used as a management tool allowing the comparison with the acceptable total risk and showing trends resulting from mitigation activity. For the purposes of this standard the risk associated with an event concerning the facilities and infrastructures of the data centre which disrupts the provision of service of the data centre is defined as event risk which is a function of impact and probability where

a) impact is the magnitude or severity of adverse incidents or impacts, expressed numerically or nominally expected duration of loss of service (availability) of the event;

b) probability is the likelihood of the event.

The impact of risk may be assessed using different units of measure e.g. cost, safety etc.

The total risk to the functional capability of the data centre is a function of the event risks associated with each facility and infrastructure provided that those risks are quantified on the same basis. If related to the output of the business impact analysis (see 5.2) the financial value of the total risk can be estimated.

The risks considered should include external threats which can affect the facilities and infrastructures including in particular the location, which could be geographical (air traffic, flooding etc.), political (wars, trouble spots, terror etc.) or affecting neighbourhood relations (if, for example, fire hazards exist due to filling stations, chemical storage etc.) and thus influence the likelihood of a potential downtime. In addition, potential risks resulting from internal and external attacks by the staff or others should be part of the overall risk evaluation.

Impact can be categorized as:

1) low: Loss of non-critical services;

2) medium: Failure of critical system functional elements but no loss of redundancy;

3) high: Loss of critical system redundancy but no loss of service;

4) critical: Loss of critical service or loss of life (which may be extended to address personal injury).

The probability of an event occurring can be defined in a similar way, that is:

1) very low;

2) low;

3) medium;

4) high.

Each risk can be quantified on a risk map as shown in Figure 2. High risk events inhabit the top right hand corner of the figure and low risk events inhabit the bottom left hand corner.

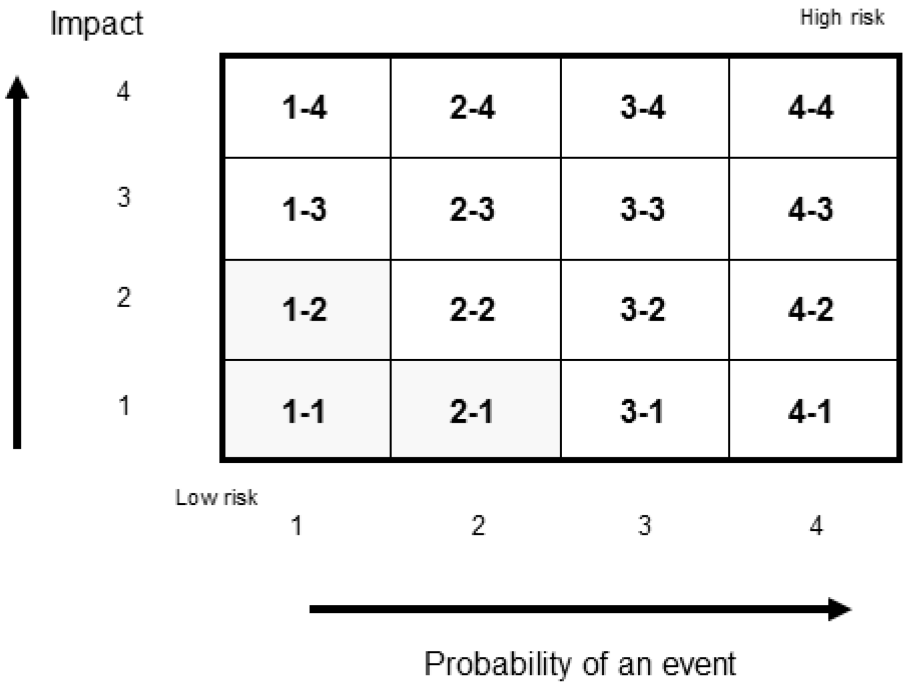

Figure 2 – Example of risk map

Having identified the risk of the possible events associated with data centre facilities and infrastructures, the downtime cost with that event shall be determined to enable design decisions to be made that reduce the risk (by means of reducing the impact or probability of the event).

6 Data centre design overview

6.1 General

Data centres differ in terms of their purpose e.g. co-hosting, co-location, enterprise, network operator services. Data centres can also differ significantly with respect to their physical size ranging from:

a) a data centre in a building housing a small quantity of storage and server equipment to provide information technology services to the occupants of that building, to

b) a data centre housing a large quantity of such equipment providing information technology services via diverse internal and external telecommunications networks and requiring sophisticated power distribution and environmental control facilities housed in one or more buildings dedicated to ensuring the operation of the data centre.

This clause provides a general design overview for data centres independent of their purpose and their size.

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6.2 Spaces and facilities

Figure 3 shows a schematic representation of the spaces required by a large data centre within a building and within premises containing one or more building.

The data centre may share certain spaces with the rest of the building including:

a) building entrance facilities;

b) personnel entrance(s);

c) docking/loading bay(s);

d) generators space(s) including fuel storage;

e) transformer space(s);

f) electrical distribution space(s);

g) telecommunications spaces(s).

The need for the above spaces and facilities within the building depends upon the purpose of both the building and the data centre. Any sharing of these spaces and facilities will depend not only on the size but also on the defined Availability and Protection Classes of the data centre and the functions of the remainder of the building. For example, in buildings housing large data centres, the facilities and spaces supporting the data centre may be dedicated to the data centre with separate spaces being provided for the remainder of the building.

The area within the building designated as a data centre may contain the following spaces:

1) personnel entrance(s);

2) main distributor space(s);

3) computer room space(s) and associated testing space(s);

4) electrical space(s);

5) mechanical space(s);

6) control room space(s);

7) office space(s);

8) storage and holding space(s).

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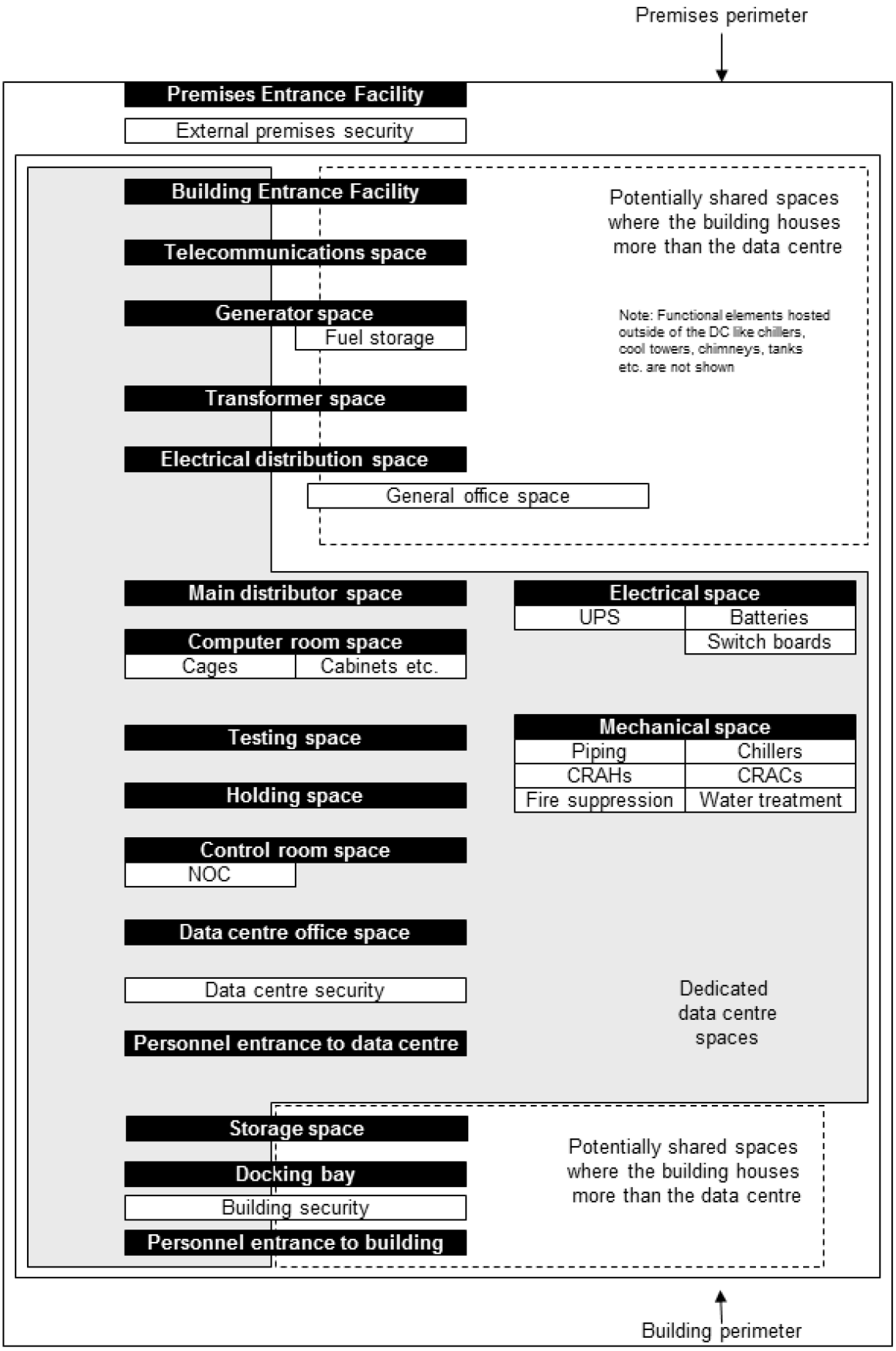

Figure 3 – Typical schematic diagram of premises containing a data centre

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Within the area of the building designated as a data centre, the need for, and contents of, the spaces depends upon the purpose of the data centre, its anticipated power consumption and the need for environmental control.

The need for segregation of spaces depends on availability and safety considerations, requirements for security and upon the need for environmental control.

As examples, a small enterprise data centre may comprise a single room having the function of a computer room space and an electrical space without physical segregation whereas a large data centre may require one or more segregated spaces of each type identified in Figure 3.

7 Classification system for the design of data centre facilities and infrastructures

7.1 General

For the purposes of the EN 50600 series, data centres facilities and infrastructures are designated with respect to:

a) Availability Classes (see 7.2);

b) Protection Classes (see 7.3);

c) Energy efficiency enablement levels (see 7.4).

These designations are used in combination to determine the relevant requirements and recommendations for the following facilities and infrastructures:

1) building construction (see EN 50600-2-1);

2) power distribution (see EN 50600-2-2);

3) environmental control (see EN 50600-2-3);

4) telecommunications cabling infrastructure (see EN 50600-2-4);

5) security systems (see EN 50600-2-5).

7.2 Availability

7.2.1 General

Data centres can be single-site or configured to operate across multiple sites.

7.2.2 describes the availability concepts and requirements for a single-site data centre and 7.2.3 describes the use of a multi-site data centre to improve the overall service availability.

7.2.2 Single-site data centres

The required availability of the facilities and infrastructures that support the functionality of the data centre is of the utmost significance. The data centre owner/user shall determine the desired availability of the overall set of facilities and infrastructures using business risk analysis and business impact analysis (Clause 5). It is recognized that availability requirements may vary with time of day, week or month.

Different qualitative Availability Classes for the overall set of data centre facilities and infrastructures are defined as shown in Table 1. The availability of the entire data centre depends on the Availability Classes of its individual infrastructures such as power sourcing and distribution, environmental control and security. The requirements for a specific facility or infrastructure of a given Availability Class are specified in the EN 50600-2-X series.

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In order for the set of facilities and infrastructures of data centre to be considered to be of a given Availability Class, the design of each individual facility and infrastructure listed in Table 1 shall meet or exceed that Availability Class.

The provision of higher Availability Classes generally requires greater investment. Additional details about availability are provided in Annex A.

Table 1 – Availability Classes and example implementations

	Availability Class 1	**Availability Class 2**	**Availability Class 3**	**Availability Class 4**
Availability of overall set of infrastructures	**low**	**medium**	**high**	**very high**
Definition for power supply (see EN 50600-2–2)	**Single path to primary distribution equipment - Single source**	**Single path to primary distribution equipment - Redundant sources**	**Multiple paths to primary distribution equipment - Redundant sources**	**Multiple paths to primary distribution equipment - Multiple sources**
Definition for power distribution (see EN 50600-2–2)	**Single path**	**Single path with redundancy**	**Multiple paths - Concurrent repair/operate solution**	**Multiple paths - Fault tolerant except during maintenance**
Definition for environmental control (see EN 50600-2–3)	**Single path**	**Single path with redundancy**	**Multiple paths - Concurrent repair/operate solution**	**Multiple paths - Fault tolerant except during maintenance**
Definition for telecommunications cabling (see EN 50600-2–4)	**Single path - direct connections or fixed infrastructure with single access network connection**	**Single path - fixed infrastructure with multiple access network connections**	**Multiple paths - fixed infrastructure with diverse pathways with multiple access network connections**	**Multiple paths - fixed infrastructure with diverse pathways and redundant distribution zones and multiple access network connections**
NOTE 1: Requirements and recommendations for data centre construction that provide the desired Protection Classes to ensure availability of the facilities and infrastructures are addressed in EN 50600-2–1. NOTE 2: Requirements and recommendations for physical security of data centre spaces and pathways to ensure availability of the facilities and infrastructures are addressed in EN 50600-2–5.				

The EN 50600 series defines four Classes of availability. Based on the outcome of the business risk analysis in Clause 5 an Availability Class shall be selected for the following infrastructures:

- power supply and distribution;
- environmental control;
- telecommunications cabling.

The selection of the Availability Class shall be made based on the following general guidance (for guidance specific to each individual infrastructure see the appropriate part of EN 50600-2-X).

A Class 1 solution (single path) is appropriate where the outcome of the risk assessment deems it acceptable that:

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- a single fault in a functional element can result in loss of functional capability;
- planned maintenance can require the load to be shut-down.

A Class 2 solution (single path with redundancy) is appropriate where the outcome of the risk assessment deems it necessary that:

- a single fault in a device shall not result in loss of functional capability of that path (via redundant devices);
- routine planned maintenance of a redundant device shall not require the load to be shut down.

NOTE Failure of the path can result in unplanned load shutdown and routine maintenance of non-redundant devices can require planned load shutdown.

A Class 3 solution (multiple paths providing a concurrent/repair operate solution) is appropriate where the outcome of the risk assessment deems it necessary that:

- a fault of a functional element shall not result in loss of functional capability;
- planned maintenance shall not require the load to be shut-down;
- for environmental control: although a failure of a path can result in unplanned load shutdown, maintenance routines shall not require planned load shutdown as the passive path serves to act as the concurrent maintenance enabler as well as reducing the recovery of service time (minimizing the mean downtime) after the failure of a path.

All paths shall be designed to sustain the maximum load.

A Class 4 solution (fault tolerant solution except during maintenance) is appropriate where the outcome of the risk assessment deems it necessary that:

- a fault of a functional element shall not result in loss of functional capability;
- planned maintenance shall not require the load to be shut-down;
- for power supply and distribution: any single event impacting a functional element shall not result in load shut-down;
- for environmental control: a failure of one path shall not result in unplanned load shutdown

All paths shall be designed to sustain the maximum load.

More information about availability can be found in Annex A.

Additional attention shall be given to the physical security of the facilities and infrastructures outlined in 7.3, describing other important factors for the overall availability of the entire data centre.

In addition to the design and installation of more sophisticated technical solutions, the implementation of higher Availability Classes implies the application of effective organisational structures to manage the operation of those technical solutions including, but not limited to:

1) the availability of trained service personnel;
2) storage of spare parts;
3) the establishment of maintenance contracts and service level agreements;
4) rapid access to precise instructions defining the actions and communications required in any case of failure.

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7.2.3 Multi-site data centres

In some cases data centres configured across multiple sites can feature individual sites of low availability class infrastructures while maintaining the overall service availability objectives provided by the group of data centre sites. This standard does not provide a mapping between the overall service availability of the multi-site data centre and that of the availability class of the infrastructures in any individual site. This concept requires additional capabilities of the IT services which are out of the scope of this document.

7.3 Physical security

7.3.1 General

Each of the data centre spaces and pathways, independent of the size or purpose of the data centre, is designated as being of a particular Protection Class. There is no concept of a data centre of a given Protection Class.

The physical security provided for the data centre has an influence on both the probability and impact of risk events (see 5.3) since the objective of physical security is to protect against:

a) unauthorised access (see 7.3.2);

b) intrusion (see 7.3.3);

c) internal environmental events (see 7.3.4);

d) external environmental events (see 7.3.4).

The required protection classes shall be selected according EN 50600-2-5 for each of these objectives. The protection classes for intrusion are for further study and are subject of the next revision of EN 50600-2-5.

7.3.2 Protection against unauthorised access

The areas of the data centre and its surroundings shall be protected against unauthorised access.

Within the data centre, the access restrictions are dependent on the purpose of the data centre (e.g. enterprise vs. co-location) and on the function of the data centre spaces and pathways. The design criteria are based upon an analysis of needs defining appropriate requirements and recommendations.

EN 50600-2-5 specifies the requirements of and provides recommendations for active and passive measures in support of the Protection Classes for unauthorized access.

EN 50600-2-1 specifies requirements and recommendation for the construction of boundaries between spaces and pathways of a given Protection Class.

7.3.3 Protection against intrusion

The areas of the data centre and its surroundings shall be protected against intrusion.

Within the data centre, the intrusion measures are dependent on the purpose of the data centre (e.g. enterprise vs. co-location) and on the function of the data centre spaces and pathways. The design criteria are based upon an analysis of needs defining appropriate requirements and recommendations.

For a particular Protection Class the intrusion delay time provided by an intrusion barrier should be longer than the time it takes to stop the intruder. If the intrusion delay is created by multiple barriers, the summation of the individual delay times results in the total intrusion delay time.

All intrusion related requirements and recommendations for the construction of data centres are for further study and subject of the next revision of EN 50600-2-1.

All intrusion related requirements and recommendations for active and passive measures in support of the Protection Classes are for further study and subject of the next revision of EN 50600-2-5.

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7.3.4 Protection against environmental events

The areas of the data centre and its surroundings shall be protected against environmental events.

Protection against internal and external environmental events includes all measures required to ensure the desired Availability Class for the facilities and infrastructures of the data centre including building construction, protection systems and organisational measures.

Internal environmental events include overheating, fire, electrostatic discharge, water etc. impacting the function of the data centre infrastructures.

External environmental events include fire, flood, earthquake, explosion and other forms of natural disaster (lightning and other electromagnetic affects).

Under optimal conditions, the risks posed by external environmental events are mitigated by the selection of the data centre location (see EN 50600-2-1). However, in most situations alternative design solutions have to be applied to the data centre facilities and infrastructures to provide them with an acceptable degree of security against such events.

EN 50600-2-2 specifies the Protection Classes applicable to spaces and pathways accommodating power supply and distribution systems.

EN 50600-2-3 specifies the Protection Classes applicable to spaces and pathways accommodating environmental control systems.

EN 50600-2-5 specifies the requirements of and provides recommendations for security and protection systems in support of the Protection Classes. EN 50600-2-1 specifies requirements and recommendations for the:

a) construction of boundaries between spaces and pathways of a given Protection Class to minimize the impact of internal environmental events;

b) location and construction of data centres to mitigate external environmental events.

7.4 Energy efficiency enablement

7.4.1 General

The ability to measure the energy consumption and to allow calculation and reporting of resource management (e.g. energy efficiency, source diversity and mix) and of the various facilities and infrastructures supporting the operation of a data centre is critical to the achievement of any related objectives.

The data centre owner/user shall define the appropriate energy efficiency enablement level prior to the data centre design.

The desired energy efficiency enablement level can be determined by:

a) an operating cost analysis;

b) the application of resource and energy management processes according to EN 50600-3-1;

c) the selection and application of one or more appropriate KPIs for resource management according to the EN 50600-4-x series;

d) external regulatory or legislative requirements;

e) user defined rules.

Three levels of granularity for the measurement are defined:

1) Level 1: a measurement regime providing simple global information for the data centre as a whole;

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2) Level 2: a measurement regime provides detailed information for specific facilities and infrastructures within the data centre;

3) Level 3: a measurement regime provides granular data for systems within the spaces and pathways of the data centre.

Moving from one granularity level to a higher level requires an increased level of measurement / monitoring infrastructure.

7.4.2 Power distribution system

EN 50600-2-2 describes the power distribution infrastructure for data centres and defines the requirements and recommendations for the measurement/monitoring infrastructures of the power distribution systems in support of the desired granularity level.

7.4.3 Environmental monitoring and control

EN 50600-2-3 describes the environmental control infrastructure for data centres and defines the requirements and recommendations for the measurement/monitoring infrastructures of the environmental control systems in support of the desired granularity level.

7.4.4 Operational processes and KPIs

EN 50600-3-1 describes processes and Key Performance Indicators (KPIs) for resource and energy management which analyse data provided by monitoring of power distribution and environmental control infrastructures. Standards in the EN 50600-4-X series specify the detailed requirements for this type of KPI.

8 Design and implementation process

8.1 General

Effective data centre design requires the splitting of the project into phases. Each phase has own input and output. All these phases follow a sequential timeline, resulting in the final project plan, leading to the issuing of a contract for the installation of the data centre enabling the operational phase to commence. Phases can be executed several times if required to achieve the agreed or defined objectives. Figure 4 lists all phases in their sequential order including phase descriptions and responsibilities.

Data Centre owners should be aware of the impact of operational strategy on data centre availability, security concept, data centre management and operation. An operational concept should be discussed and decided to ensure that room layouts and security zones and protection class boundaries provide the necessary function to protect the data centre against unauthorized access according to the security concept.

The operational concept should also describe process interfaces between owner, operator, customers and suppliers. Processes, roles and responsibilities need to be defined prior to beginning of operation. Operational staff needs to be instructed on technical infrastructure and trained on operational procedures latest during acceptance test phase (see EN 50600-3-1 for more information about acceptance test).

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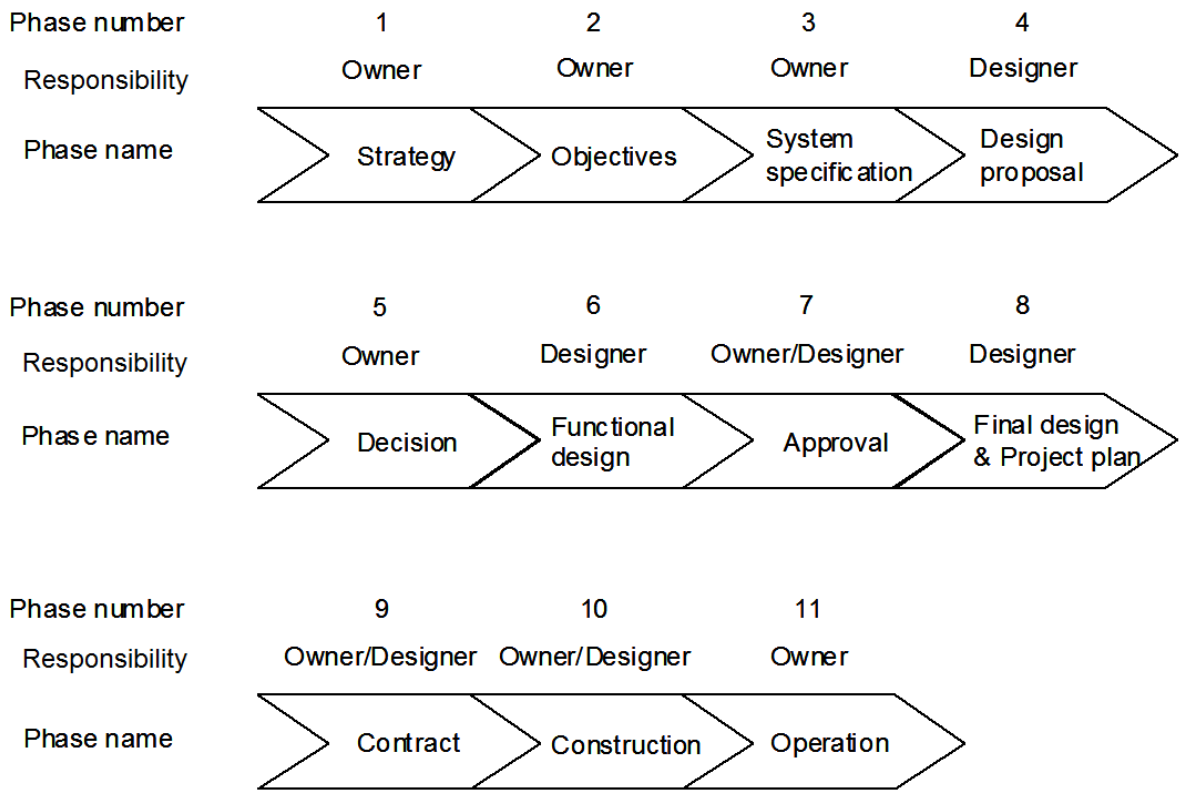

Figure 4 – Design phases

Between steps 6 and 7, an assessment shall verify that the design, the operation and management processes and the KPIs meets the project objectives. CLC/TR 50600-99-3 shall be used for these assessment purposes.

8.2 Design phases

8.2.1 Phase 1 - Strategy

This phase is for information collection in order to define the project objectives. The following information is required:

a) business risk analysis;

b) IT strategy;

c) corporate data centre strategy;

d) general customer requirements/expectations;

e) analysis of current load/demand/costs;

f) expected infrastructure technology roadmap;

g) “forecast” of future facility and infrastructure demand (space, power and locations);

h) operational strategy.

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8.2.2 Phase 2 - Objectives

This phase is used by the owner to convert the strategy into objectives. The results are the following:

a) correlation with corporate data centre strategy;

b) design benchmarks (e.g. size/performance levels/budgets);

c) project risk analysis (internal and external);

d) selecting location options

e) definition of work flow;

f) timelines and impact of delays;

g) general floor plan and materials catalogue.

8.2.3 Phase 3 - System specifications

This phase defines the target specifications for all infrastructures with the following output:

a) target specification for power supply (sources);

b) target specification for power distribution;

c) target specification for environmental control;

d) target specification for physical security;

e) target specification for fire detection and fire fighting;

f) target specifications for telecommunications infrastructure;

g) target specification for data centre operation and management;

h) target specification for construction.

8.2.4 Phase 4 - Design proposal

The designer uses the target specifications and objectives to create a design proposal for all infrastructures offering several options to the owner. The design proposal contains:

a) design proposal for power supply (sources);

b) design proposal for power distribution;

c) design proposal for environmental control;

d) design proposal for physical security;

e) design proposal for fire detection and fire fighting;

f) design proposal specifications for telecommunications infrastructure;

g) design proposal for data centre operation and management;

h) design proposal for construction;

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i) cost models and timelines for proposed options.

j) final location selection.

8.2.5 Phase 5 - Decision

The owner selects the design from the available design options and cost models (supported by designer).

8.2.6 Phase 6 - Functional design

The designer converts the owner selection into functional design. The functional design contains:

a) functional design for power supply (sources);

b) functional design for power distribution;

c) functional design for environmental control;

d) functional design for physical security;

e) functional design for fire detection and fire fighting;

f) functional design for telecommunications infrastructure;

g) functional design for data centre operation and management;

h) functional design for construction;

i) cost model “fine tuning” for selected option.

8.2.7 Phase 7 - Approval

The owner approves the functional design and cost models taking into account the risks and scheduling constraints of the project.

8.2.8 Phase 8 - Final design and project plan

The designer defines volume and/or pieces for all the infrastructures approved under 8.2.7. Furthermore, the project workflow and all project milestones and timelines are defined and subject to change control, resulting in an overall implementation plan.

8.2.9 Phase 9 - Contract

The owner (with support of the designer/consultant) issues a tender and selects the contractor(s).

8.2.10 Phase 10 - Construction

The owner and/or the designer supervise the construction over the entire construction time. Acceptance verification (testing and commissioning) for all infrastructures and for the entire data centre is executed until the data centre is put into service. Further details on testing and commissioning can be found in EN 50600-3-1.

8.2.11 Phase 11 - Operation

Hand over to owner for operation. For further details see EN 50600-3-1.

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9 Design Principles

9.1 Design reference documentation

The outcome of the steps of Clauses 5 to 7 shall be collected in a design reference document which contains as a minimum:

a) the outcome of business impact analysis in accordance with 5.2;

b) the outcome of risk analysis in accordance with 5.3;

c) the description of a base data centre strategy and the selected Availability Class according to 7.2 selected using a business risk analysis;

d) the application of physical security in accordance with 7.3;

e) the selection of energy efficiency enablement level in accordance with 7.4;

f) the operational concept in accordance with 8.1.

9.2 Design principles to support energy efficiency

The design of data centres shall consider energy efficiency (and wider aspects of resource efficiency) as a principle objective independent of the Availability Class to be applied.

EN 50600-4-X provides a series of Key Performance Indicators some of which can be employed at the design stage to assess the predicted energy efficiency.

Recommended practices about energy efficient data centre design and more information about resource and energy efficiency in data centres can be found in the following documents:

CLC/TR 50600-99-1;

CLC/TR 50600-99-2;

ETSI TS 105 174-2-2.

9.3 Design principles for EMI

The design of data centres shall consider EMI as a principle objective. Additional information can be found in

EN 50600-2-1,

EN 50600-2-2,

EN 50600-2-4,

EN 50600-2-5.

9.4 Design principles to support operational excellence

The design of data centres shall consider operational excellence as a principle objective independent of the Availability Class to be applied.

The design of data centres should enable the provision of management and operational information required by EN 50600-3-1.

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Annex A
(informative)
Overall availability and infrastructure availability

This annex explains the availability concept of EN 50600 series and highlights the differences to other concepts used in the market place.

The availability A of a data centre in operation can be calculated using Formula (A.1). For comparable reasons, the observation time span should be 1 year which equals 8760 h.

$$A = \frac{t_{\text{up}}}{t_{\text{up}} + t_{\text{down}}} \tag{A.1}$$

where

t_{up} uptime

t_{down} downtime

The availability can be described in percentage terms (see Table A.1).

Table A.1 — Availability and annual downtime

Availability A	Common reference	Downtime (based on 8760 h per year)
90 %	1-nine	36,5 days
99 %	2-nines	3,65 days
99,9 % (3-nines)	3-nines	8,76 h
99,99 % (4-nines)	4-nines	52,6 min
99,999 % (5-nines)	5-nines	5,3 min
99,9999 % (6-nines)	6-nines	31,5 s

When applied to the overall availability of a data centre, such a percentage objective based on Formula (A.1) can be valid.

However, a percentage approach is not applicable to describe objectives for the infrastructures required to support that overall availability. A discussion of availability in terms uptime and downtime hides some fundamental problems for a given infrastructure. It is necessary to introduce the concept of mean time between failure (MTBF) and the mean time to repair (MTTR) following that failure. This is highlighted in the example below.

As shown in Table A.1, an infrastructure with an availability of 99,99 % allows for approximately 53 min downtime per annum. While this might be a single 53 min period of downtime with an MTBF of approximately 365 days, it could equally be 53 separate occurrences of one minute periods of downtime with a MTBF of about one week (or even 3150 separate one second periods of downtime with an MTBF of approximately 3 h).

A failure of power supply to ICT equipment of 1 s would almost certainly result in shutdown of the equipment. The additional time for the data centre to recover its intended functionality for data processing, storage, and transport would depend on a number of factors which would render the overall availability to be substantially lower than 99,99 %. This is shown in Figure A.1 for 10, 100 and 1000 periods of downtime.

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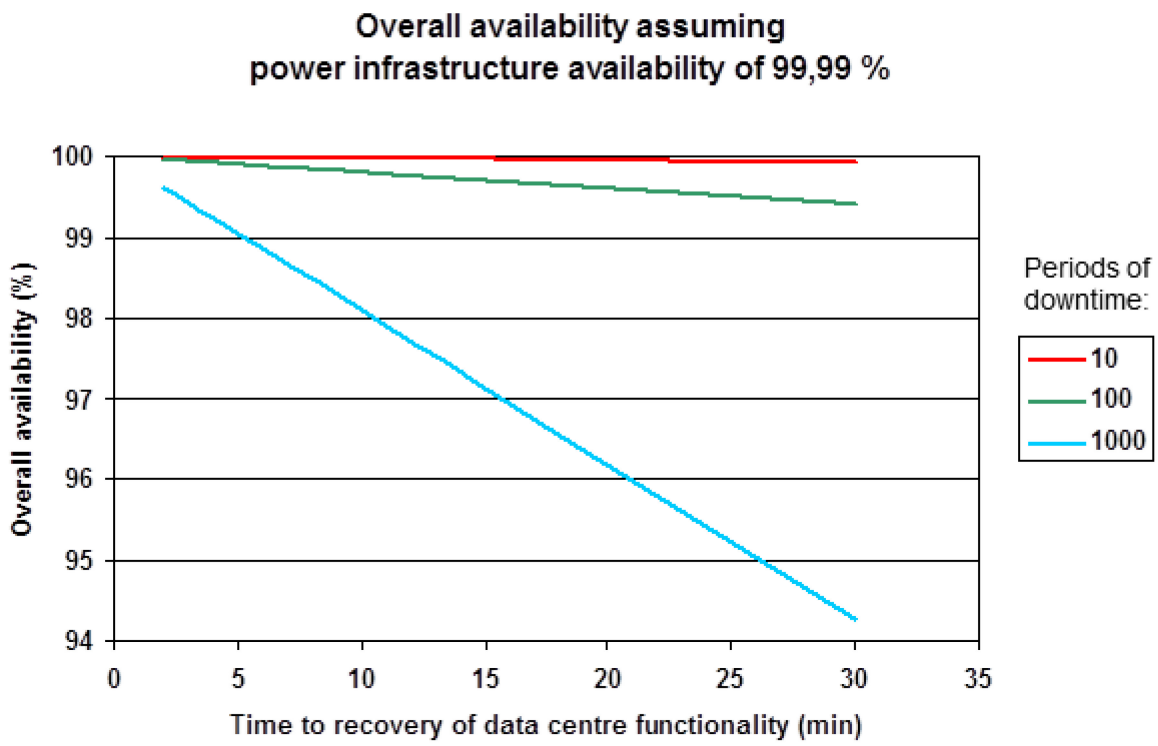

Figure A.1 — Impact of recovery of functionality on overall availability

Reversing this calculation for a given infrastructure, the achievement of a given overall availability would not only require an infrastructure availability many times better than the overall availability but would also place demands on the number of periods of downtime and the time to recovery of functionality which has no relevance to the infrastructure.

For example, it is assumed that the average time to recovery following an individual period of downtime is 10 min; as a consequence, in order for the overall availability to be 99,99 % as shown in Table A.1 the following applies:

a) if there were 2 incidents per year, the infrastructure availability would have to be 99,9934 %;

b) if there were 5 incidents per year, the infrastructure availability would have to be 99,9995 %.

For a given infrastructure it can be better to consider the modified Formula (A.2) for intrinsic availability, A_i,

$$A_i = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}} \qquad \text{(A.2)}$$

where the inherent availability describes the availability provided by the design under ideal conditions of operation and maintenance.

Using Formula (A.2) it can be seen that an infrastructure with a longer MTBF (which means a lower frequency of failures) will have a higher availability. Conversely the MTTR has the opposite effect.

However, it should be noted that the demands for MTBF and MTTR vary substantially for different infrastructures.

In consequence of the example above, the power supply (EN 50600-2-2) and the environmental control (EN 50600-2-3) have been taken into account to determine the overall availability of a data centre. Availability of the power supply and distribution system needs to address events in the sub-second (or even millisecond) range to avoid ICT equipment shut-down. In extreme cases, a sub-second failure of the

power supply and/or distribution system could potentially result in an overall data centre service downtime of several hours or even days.

By comparison, the environmental control system (EN 50600-2-3) would typically tolerate failures of a minute (or multi-minute) range without any effect on overall data centre availability. Therefore the use of a single value for infrastructure availability is non-viable without qualification.

In Clause 7, the four availability classifications of low, medium, high, and very high are defined on a more abstract level. By this qualitative classification, structural requirements and redundancy considerations can be taken into account. Annex B presents an overview in tabular form.

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Annex B
(informative)
Availability description

Table B.1 gives more details about the availability classification of this standard.

Table B.1 – Summary of availability classification

item	Availability Class			
	Class 1	Class 2	Class 3	Class 4
EN 50600-2–2 Power Supply				
Availability	Low	Medium	High	Very high
Redundant sources	N	Y	Y	Y
Protected against source failure	N	Y	Y	Y
Redundant path to primary distribution	N	N	Y	Y
Protected against path failure	N	N	Y	Y
Compartmentalization	N	N	N	Y
Protected against single device failure	N	Y	Y	Y
Load operation during maintenance	N	N[1)]	Y	Y
Fault tolerant	N	N	Y[2)]	Y
EN 50600-2–2 Power Distribution				
Availability	Low	Medium	High	Very high
Redundant path	N	N	Y	Y
Protected against path failure	N	N	Y	Y
Compartmentalization	N	N	N	Y
Protected against single device failure	N	Y	Y	Y
Load operation during maintenance	N	N[1)]	Y	Y
Fault tolerant	N	N	N	Y[2)]
EN 50600-2–3 Environmental Control				
Availability	Low	Medium	High	Very high
Redundant source	N	N	Y	Y
Redundant path	N	N	Y	Y
Protected against path failure	N	N	Y	Y
Compartmentalization	N	N	N	Y
Protected against single device failure	N	Y	Y	Y
Load operation during maintenance	N	N[1)]	Y	Y
Fault tolerant	N	N	N	Y[2)]

[1)] Depending on the device being maintained.

[2)] Except during maintenance.

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EN 50173-5:2018, *Information technology - Generic cabling systems - Part 5: Data centre spaces*

EN 50174-1:2018, *Information technology - Cabling installation - Part 1: Installation specification and quality assurance*

EN 50174-2:2018, *Information technology - Cabling installation - Part 2: Installation planning and practices inside buildings*

EN 50310, *Telecommunications bonding networks for buildings and other structures*

EN 62040-1:201X, *Uninterruptible power systems (UPS) - Part 1: Safety requirements*

EN 62305 (all parts), *Protection against lightning (IEC 62305 (all parts))*

EN ISO 22301, *Societal security - Business continuity management systems - Requirements (ISO 22301)*

CLC/TR 50600-99-1, *Information technology - Data centre facilities and infrastructures - Part 99-1: Recommended practices for energy management*

ETSI TS 105 174-2-2, *Broadband Deployment – Energy Efficiency and Key Performance Indicators; Part 2: Network sites; Sub-part 2: Data centres*

IEC 60050-191, *International Electrotechnical Vocabulary – Part 191: Dependability and quality of service*

ISO/TS 22317, *Societal security — Business continuity management systems — Guidelines for business impact analysis (BIA)*

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